

Unit 4 Geometry

4.1 Volume and Area

1

★ Mark for Review

ABC

Square A has side lengths that are **166** times the side lengths of square B. The area of square A is k times the area of square B. What is the value of k ?

2

★ Mark for Review

ABC

A cube has an edge length of 68 inches. A solid sphere with a radius of 34 inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube *not* taken up by the sphere?

A. 149,796

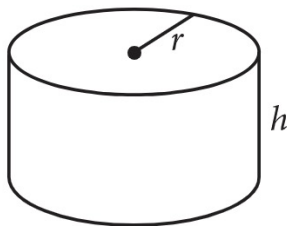
B. 164,500

C. 190,955

D. 310,800

3

★ Mark for Review

ABC

The figure shown is a right circular cylinder with a radius of r and height of h . A second right circular cylinder (not shown) has a volume that is 392 times as large as the volume of the cylinder shown. Which of the following could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder?

A. $R = 8r$ and $H = 7h$

B. $R = 8r$ and $H = 49h$

C. $R = 7r$ and $H = 8h$

D. $R = 49r$ and $H = 8h$

4

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ABC

A cube has a volume of 474,552 cubic units. What is the surface area, in square units, of the cube?

5

★ Mark for Review

ABC

A right circular cone has a height of 22 centimeters (cm) and a base with a diameter of 6 cm. The volume of this cone is $n\pi \text{ cm}^3$. What is the value of n ?

6

★ Mark for Review

ABC

A manufacturer determined that right cylindrical containers with a height that is 4 inches longer than the radius offer the optimal number of containers to be displayed on a shelf. Which of the following expresses the volume, V , in cubic inches, of such containers, where r is the radius, in inches?

A. $V = 4\pi r^3$

B. $V = \pi(2r)^3$

C. $V = \pi r^2 + 4\pi r$

D. $V = \pi r^3 + 4\pi r^2$

7

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ABC

A rectangular poster has an area of 360 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 20%. What is the area of the copy, in square inches?

8

★ Mark for Review

ABC

A right rectangular prism has a length of **28 centimeters (cm)**, a width of **15 cm**, and a height of **16 cm**. What is the surface area, in cm^2 , of the right rectangular prism?

9

★ Mark for Review

ABC

A right circular cone has a volume of $\frac{1}{3}\pi$ cubic feet and a height of **9 feet**. What is the radius, in feet, of the base of the cone?

A. $\frac{1}{3}$

B. $\frac{1}{\sqrt{3}}$

C. $\sqrt{3}$

D. 3

10

★ Mark for Review

ABC

Two identical rectangular prisms each have a height of 90 centimeters (cm). The base of each prism is a square, and the surface area of each prism is K cm². If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K$ cm². What is the side length, in cm, of each square base?

A. 4

B. 8

C. 9

D. 16

11

★ Mark for Review

ABC

The volume of right circular cylinder A is 22 cubic centimeters. What is the volume, in cubic centimeters, of a right circular cylinder with twice the radius and half the height of cylinder A ?

A. 11

B. 22

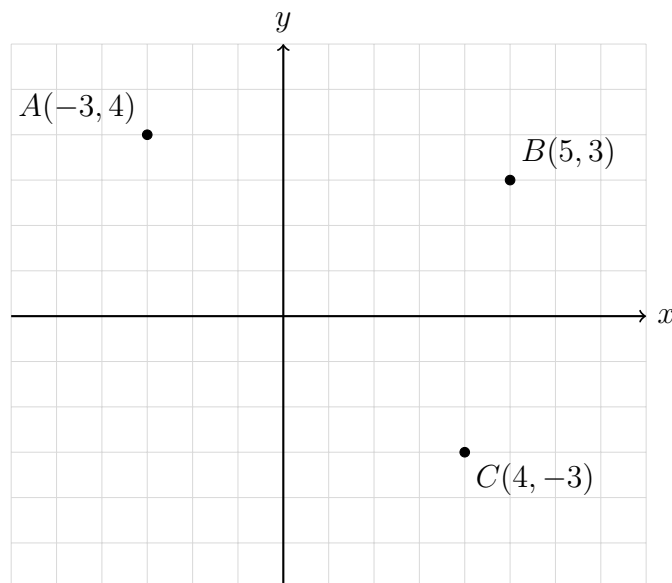
C. 44

D. 66

12

★ Mark for Review

ABC



What is the area, in square units, of the triangle formed by connecting the three points shown?

13

★ Mark for Review

ABC

Circle A has a radius of $3n$ and circle B has a radius of $129n$, where π is a positive constant. The area of circle B is how many times the area of circle A ?

A. 43

B. 86

C. 129

D. 1,849

14

★ Mark for Review

ABC

The dimensions of a right rectangular prism are 4 inches by 5 inches by 6 inches. What is the surface area, in square inches, of the prism?

A. 30

B. 74

C. 120

D. 148

15

★ Mark for Review

ABC

A right triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. What is the area of the triangle, in square units?

A. $8\sqrt{2} + \sqrt{80}$

B. 12

C. $24\sqrt{80}$

D. 24

16

★ Mark for Review

ABC

Triangles ABC and DEF are similar. Each side length of triangle ABC is 4 times the corresponding side length of triangle DEF . The area of triangle ABC is 270 square inches. What is the area, in square inches, of triangle DEF ?

17

★ Mark for Review

ABC

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

A. 18

B. 27

C. 36

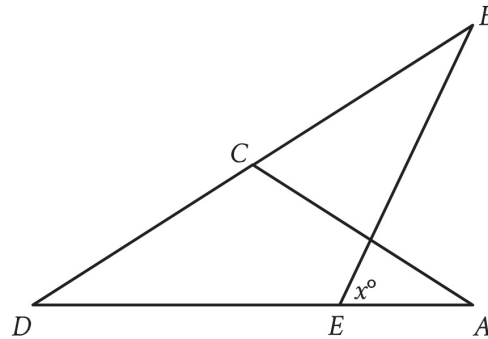
D. 81

4.2 Line, Angles, and Triangles

1

★ Mark for Review

ABC



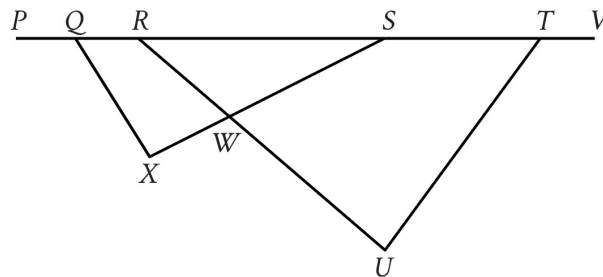
Note: Figure not drawn to scale.

In the figure, $AC = CD$. The measure of angle $\angle EBC$ is 45° , and the measure of angle $\angle ACD$ is 104° . What is the value of x ?

2

★ Mark for Review

ABC

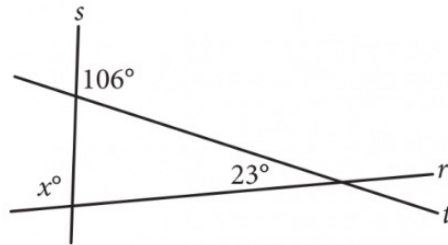


Note: Figure not drawn to scale.

In the figure shown, points Q , R , S , and T lie on line segment $\overline{PV}$, and line segment $\overline{RU}$ intersects line segment $\overline{SX}$ at point W . The measure of $\angle SQX$ is 48° , the measure of $\angle SXQ$ is 86° , the measure of $\angle SWU$ is 85° , and the measure of $\angle VTU$ is 162° . What is the measure, in degrees, of $\angle TUR$?

3

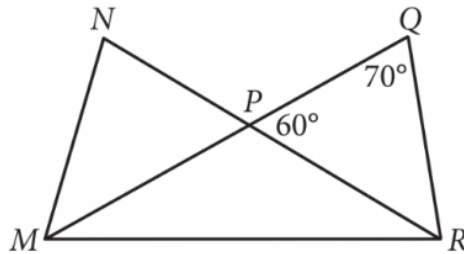
★ Mark for Review

ABC

Intersecting lines r , s , and t are shown below. What is the value of x ?

4

★ Mark for Review

ABC

In the figure above, $\overline{MQ}$ and $\overline{NR}$ intersect at point P , $NP = QP$, and $MP = PR$. What is the measure, in degrees, of $\angle QMR$? (Disregard the degree symbol when gridding your answer.)

5

★ Mark for Review

ABC

In triangle ABC , the measure of angle B is 90° and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 15 and the length of $\overline{AC}$ is 23 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

A. $\frac{15}{38}$

B. $\frac{15}{23}$

C. $\frac{23}{15}$

D. $\frac{38}{15}$

6

★ Mark for Review

ABC

In triangles ABC and DEF , angles B and E each have measure 27° and angles C and F each have measure 41° . Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF ?

A. The measure of angle A

B. The length of side $\overline{AB}$

C. The lengths of sides $\overline{BC}$ and $\overline{EF}$

D. No additional information is necessary.

7

★ Mark for Review

ABC

In triangles LMN and RST , angles L and R each have measure 60° , $\overline{LN} = 10$, and $\overline{RT} = 30$. Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST ?

A. $\overline{MN} = 7$ and $\overline{ST} = 7$

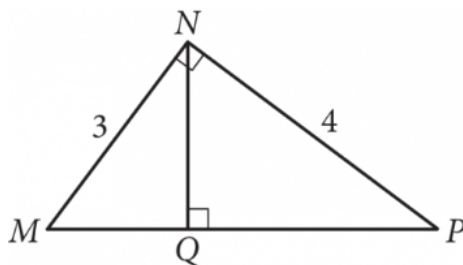
B. $\overline{MN} = 7$ and $\overline{ST} = 21$

C. The measures of angles M and S are 70° and 60° , respectively.

D. The measures of angles M and T are 70° and 50° , respectively.

8

★ Mark for Review

ABC

In the figure above, what is the length of $\overline{NQ}$?

A. 2.2

B. 2.3

C. 2.4

D. 2.5

9

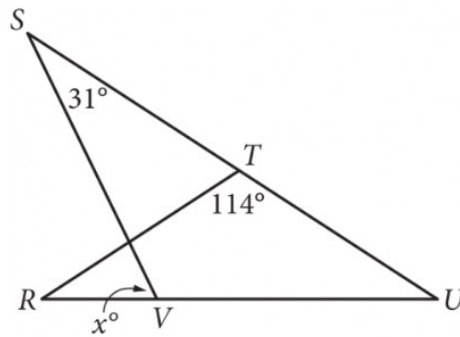
★ Mark for Review

ABC

Triangle XYZ is similar to triangle RST such that X, Y , and Z correspond to R, S , and T , respectively. The measure of $\angle Z$ is 20° and $2\overline{XY} = \overline{RS}$. What is the measure of $\angle T$?

A. 2° B. 10° C. 20° D. 40° **10**

★ Mark for Review

ABC

In the figure above, $\overline{RT} = \overline{TU}$. What is the value of x ?

A. 72

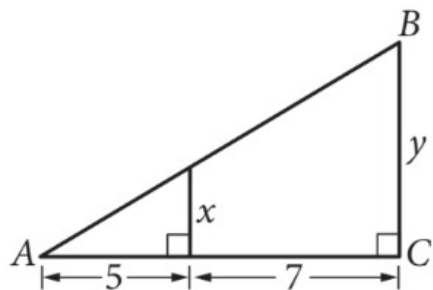
B. 66

C. 64

D. 58

11

★ Mark for Review

ABC

The area of triangle ABC above is at least 48 but no more than 60. If y is an integer, what is one possible value of x ?

12

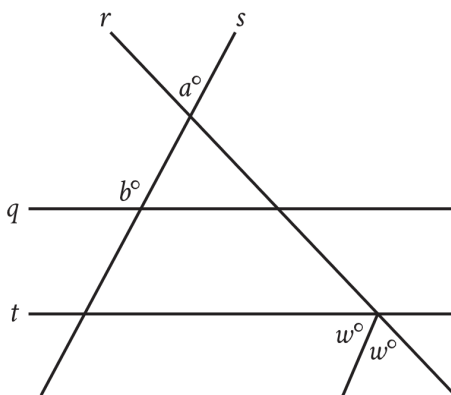
★ Mark for Review

ABC

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

13

★ Mark for Review

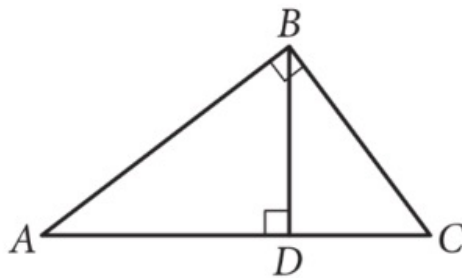
ABC

In the figure, parallel lines q and t are intersected by lines r and s . If $a = 43^\circ$ and $b = 122^\circ$, what is the value of w ?

14

★ Mark for Review

ABC



In the figure above, $\overline{BD} = 6$ and $\overline{AD} = 8$. What is the length of $\overline{DC}$?

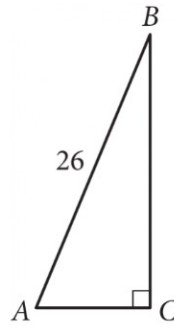
4.3 Right Triangles and Trigonometry

1

★ Mark for Review

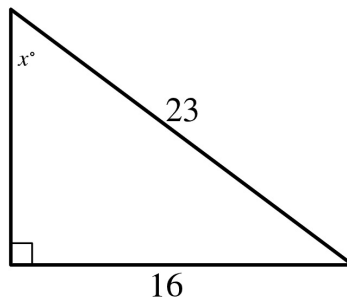
ABC

Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

**2**

★ Mark for Review

ABC

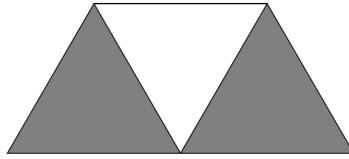


In the triangle shown, what is the value of $\sin x^\circ$?

3

★ Mark for Review

ABC



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

A. $2\sqrt{3}$ B. $4\sqrt{3}$ C. $8\sqrt{3}$

D. 16

4

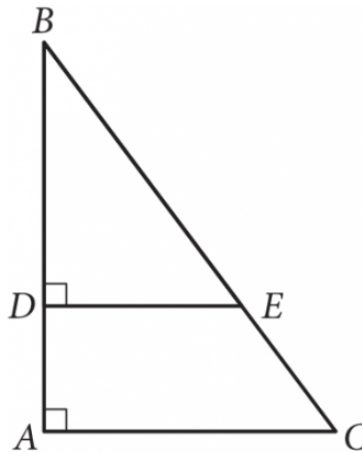
★ Mark for Review

ABC

The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k ?

5

★ Mark for Review

ABC

In the figure above, $\tan B = \frac{3}{4}$. If $BC = 15$ and $DA = 4$, what is the length of $\overline{DE}$?

6

★ Mark for Review

ABC

An isosceles right triangle has a hypotenuse of length 58 inches. What is the perimeter, in inches, of this triangle?

A. $29\sqrt{2}$

B. $58\sqrt{2}$

C. $58 + 58\sqrt{2}$

D. $58 + 116\sqrt{2}$

7

★ Mark for Review

ABC

An isosceles right triangle has a perimeter of $94 + 94\sqrt{2}$ inches. What is the length, in inches, of one leg of this triangle?

A. 47

B. $47\sqrt{2}$

C. 94

D. $94\sqrt{2}$ **8**

★ Mark for Review

ABC

The side lengths of right triangle RST are given.

$$RS = 440$$

$$ST = 384$$

$$TR = 584$$

Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

A. $\frac{48}{73}$ B. $\frac{55}{73}$ C. $\frac{48}{55}$ D. $\frac{55}{48}$

9

★ Mark for Review

ABC

A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?

10

★ Mark for Review

ABC

The side lengths of right triangle RST are given.

$$RS = 20$$

$$ST = 48$$

$$TR = 52$$

Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

A. $\frac{5}{13}$

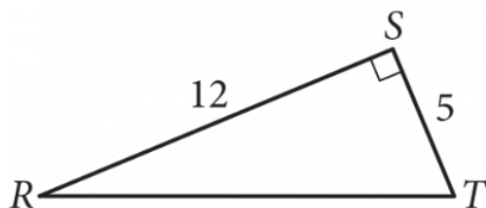
B. $\frac{5}{12}$

C. $\frac{12}{13}$

D. $\frac{12}{5}$

11

★ Mark for Review

ABC

In triangle RST above, point W (not shown) lies on $\overline{RT}$. What is the value of $\cos(\angle RSW) - \sin(\angle WST)$?

12

★ Mark for Review

ABC

Triangle ABC is similar to triangle DEF , where A corresponds to D and C corresponds to F . Angles C and F are right angles. If $\tan(A) = \sqrt{3}$ and $DF = 125$, what is the length of DE ?

A. $125\frac{\sqrt{3}}{3}$

B. $125\frac{\sqrt{3}}{2}$

C. $125\sqrt{3}$

D. 250

13

★ Mark for Review

ABC

The length of a rectangle's diagonal is $3\sqrt{17}$, and the length of the rectangle's shorter side is 3. What is the length of the rectangle's longer side?

14

★ Mark for Review

ABC

A square is inscribed in a circle. The radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. What is the side length, in inches, of the square?

A. 20

B. $\frac{20\sqrt{2}}{2}$ C. $20\sqrt{2}$

D. 40

15

★ Mark for Review

ABC

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles C and F are right angles. The length of AB is 2.9 times the length of DE . If $\tan A = \frac{21}{20}$, what is the value of $\sin D$?

16

★ Mark for Review

ABC

In a right triangle, the tangent of one of the two acute angles is $\frac{\sqrt{3}}{3}$. What is the tangent of the other acute angle?

A. $-\frac{\sqrt{3}}{3}$

B. $-\frac{3}{\sqrt{3}}$

C. $\frac{\sqrt{3}}{3}$

D. $\frac{3}{\sqrt{3}}$

17

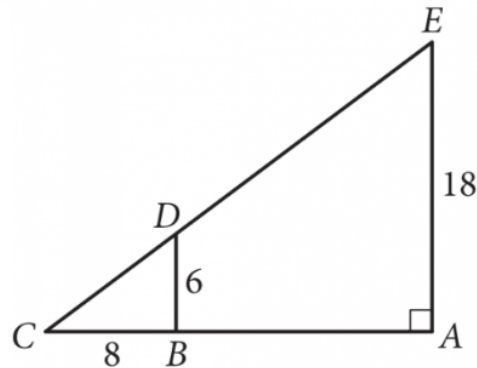
★ Mark for Review

ABC

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angle C corresponds to angle F . Angles C and F are right angles. If $\tan(A) = \frac{50}{7}$, what is the value of $\tan(E)$?

18

★ Mark for Review

ABC

In the figure above, $\overline{BD}$ is parallel to $\overline{AE}$. What is the length of $\overline{CE}$?

19

★ Mark for Review

ABC

In triangle JKL , $\cos(K) = \frac{24}{51}$ and angle J is a right angle. What is the value of $\cos(L)$?

20

★ Mark for Review

ABC

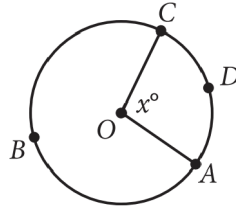
A triangle with angle measures 30° , 60° , and 90° has a perimeter of $18 + 6\sqrt{3}$. What is the length of the longest side of the triangle?

4.4 Circles

1

★ Mark for Review

ABC



The circle above has center O , the length of arc ADC is 5π , and $x = 100$. What is the length of arc ABC ?

A. 9π B. 13π C. 18π D. $\frac{13\pi}{2}$ **2**

★ Mark for Review

ABC

The graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy -plane is a circle. What is the length of the circle's radius?

3

★ Mark for Review

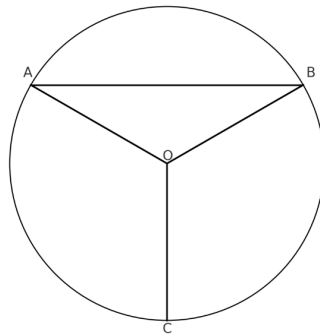
ABC

$$x^2 + 20x + y^2 + 16y = -20$$

The equation above defines a circle in the xy -plane. What are the coordinates of the center of the circle?

A. $(-20, -16)$ B. $(-10, -8)$ C. $(10, 8)$ D. $(20, 16)$ **4**

★ Mark for Review

ABC

Point O is the center of the circle above, and the measure of $\angle OAB$ is 30° . If the length of $\overline{OC}$ is 18, what is the length of arc AB ?

A. 9π B. 12π C. 15π D. 18π

5

★ Mark for Review

ABC

A circle in the xy -plane has a diameter with endpoints $(2, 4)$ and $(2, 14)$. An equation of this circle is $(x - 2)^2 + (y - 9)^2 = r^2$, where r is a positive constant. What is the value of r ?

6

★ Mark for Review

ABC

Circle A in the xy -plane has the equation $(x + 5)^2 + (y - 5)^2 = 4$. Circle B has the same center as circle A . The radius of circle B is two times the radius of circle A . The equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. What is the value of k ?

7

★ Mark for Review

ABC

What is the diameter of the circle in the xy -plane with equation $(x - 5)^2 + (y - 3)^2 = 16$?

A. 4

B. 8

C. 16

D. 32

8

★ Mark for Review

ABC

Point O is the center of a circle. The measure of arc RS on this circle is 100° . What is the measure, in degrees, of its associated angle $\angle ROS$?

9

★ Mark for Review

ABC

The equation $(x+6)^2 + (y+3)^2 = 121$ defines a circle in the xy -plane. What is the radius of the circle?

10

★ Mark for Review

ABC

A circle in the xy -plane has its center at $(-4, -6)$. Line k is tangent to this circle at the point $(-7, -7)$. What is the slope of line k ?

A. -3 B. $-\frac{1}{3}$ C. $\frac{1}{3}$ D. 3 **11**

★ Mark for Review

ABC

A circle has center O , and points R and S lie on the circle. In triangle ORS , the measure of $\angle ROS$ is 88° . What is the measure of $\angle RSO$, in degrees? (Disregard the degree symbol when entering your answer.)

12

★ Mark for Review

ABC

In the xy -plane, the graph of $2x^2 - 6x + 2y^2 + 2y = 45$ is a circle. What is the radius of the circle?

A. 5

B. 6.5

C. $\sqrt{40}$ D. $\sqrt{50}$ **13**

★ Mark for Review

ABC

The equation $x^2 + (y - 1)^2 = 49$ represents circle A . Circle B is obtained by shifting circle A down 2 units in the xy -plane. Which of the following equations represents circle B ?

A. $(x + 2)^2 + (y - 1)^2 = 49$ B. $x^2 + (y - 3)^2 = 49$ C. $(x - 2)^2 + (y - 1)^2 = 49$ D. $x^2 + (y + 1)^2 = 49$

14

★ Mark for Review

ABC

A circle has center O , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 3 inches. What is the circumference, in inches, of the circle?

A. 3

B. 6

C. 9

D. 24

15

★ Mark for Review

ABC

Which of the following equations represents a circle in the xy -plane that intersects the y -axis at exactly one point?

A. $(x - 8)^2 + (y - 8)^2 = 16$

B. $(x - 8)^2 + (y - 4)^2 = 16$

C. $(x - 4)^2 + (y - 9)^2 = 16$

D. $x^2 + (y - 9)^2 = 16$

16

★ Mark for Review

ABC

In the xy -plane, the graph of the equation above is a circle. Point P is on the circle and has coordinates $(10, -5)$. If $\overline{PQ}$ is a diameter of the circle, what are the coordinates of point Q ?

$$(x - 6)^2 + (y + 5)^2 = 16$$

A. $(2, -5)$ B. $(6, -1)$ C. $(6, -5)$ D. $(6, -9)$ **17**

★ Mark for Review

ABC

A circle in the xy -plane has its center at $(-5, 2)$ and has a radius of 9. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?

18

★ Mark for Review

ABC

The equation $x^2 + (y - 2)^2 = 36$ represents circle A . Circle B is obtained by shifting circle A down 4 units in the xy -plane. Which of the following equations represents circle B ?

A. $x^2 + (y - 6)^2 = 36$ B. $x^2 + (y + 2)^2 = 36$ C. $(x - 4)^2 + (y - 2)^2 = 36$ D. $(x + 4)^2 + (y - 2)^2 = 36$

19

★ Mark for Review

ABC

Points A and B lie on a circle with radius 1, and arc AB has length $\frac{\pi}{3}$. What fraction of the circumference of the circle is the length of arc AB ?

20

★ Mark for Review

ABC

A circle in the xy -plane has equation $(x + 3)^2 + (y - 1)^2 = 25$. Which of the following points does NOT lie in the interior of the circle?

A. $(-7, 3)$

B. $(-3, 1)$

C. $(0, 0)$

D. $(3, 2)$